ConlaneX — Clane 1 Engine

Technical Design Document v0.1

Conlane Corporation Pvt Ltd | Bengaluru, Karnataka, India | June 2026
Founder: Shashank D V

1. EXECUTIVE SUMMARY

ConlaneX is developing the Clane engine series — high-performance, reusable liquid rocket engines designed from first principles using computational physics and open-source simulation tools.

Clane 1 is a full-flow staged combustion LOX/CH4 engine targeting 2,500 kN vacuum thrust, 363 s vacuum Isp, 350 bar chamber pressure, and 227:1 thrust-to-weight ratio. These metrics meet or exceed the SpaceX Raptor 3 in thrust and TWR, at a projected production cost 60-80% lower.

Current status: TRL 2 — Physics model complete. 3D CAD assembly complete. CFD simulation initiated.
GitHub: github.com/memorareach-code/conlanex-clane1

2. MISSION AND VISION

ConlaneX's mission is to make humanity multiplanetary by building the most cost-effective, high-performance rocket propulsion systems on Earth.

The Clane engine series targets three simultaneous advantages over current state-of-the-art:
- Higher performance than existing engines in its thrust class
- Lower cost through Indian manufacturing and additive manufacturing
- Higher reusability through clean-burning methane and robust design margins

3. CLANE 1 — LOCKED SPECIFICATION

Propellants: LOX / CH4
Engine cycle: Full-Flow Staged Combustion (FFSC)
Vacuum thrust: 2,500 kN
Sea-level thrust: 2,300 kN
Vacuum Isp: 363 s
Sea-level Isp: 327 s
Chamber pressure: 350 bar
Mixture ratio O/F: 3.55
Expansion ratio: 35:1

Throat diameter: 212 mm
Nozzle exit diameter: 1,257 mm
Chamber diameter: 600 mm
Total mass flow: 702.3 kg/s
LOX mass flow: 547.9 kg/s
CH4 mass flow: 154.3 kg/s
Turbopump power: 57,500 kW (~77,000 hp)
Engine dry mass: ~1,100 kg
Thrust-to-weight ratio: ~227:1
Target reuse life: 50+ flights

4. COMPETITIVE BENCHMARK

Merlin 1D: LOX/RP-1 | Gas Generator | 97 bar | 981 kN vacuum | 311 s Isp | ~180 TWR | ~$400k
Clane 1: LOX/CH4 | FFSC | 350 bar| 2,500 kN | 363 s Isp | ~227 TWR | ~$132-200k
Raptor 3: LOX/CH4 | FFSC | 350 bar| ~2,940 kN | ~355 s Isp| ~200 TWR | ~$1M+

Clane 1 beats Raptor 3: Isp vacuum (+8s), TWR (+13.5%), production cost (-80%)

5. THERMODYNAMIC ANALYSIS

Ve = Isp x g0 = 363 x 9.80665 = 3,560 m/s
m_dot = F / Ve = 2,500,000 / 3,560 = 702.3 kg/s
m_dot_CH4 = 154.3 kg/s | m_dot_LOX = 547.9 kg/s
Throat area A* = 0.03530 m2 | D_throat = 0.212 m
Nozzle exit A = 1.2355 m2 | D_exit = 1.257 m
Turbopump power = 57,500 kW total
Gamma=1.214 | Tc=3533K | Mw=22.2 g/mol

6. ENGINE ARCHITECTURE — FFSC CYCLE

547.9 kg/s LOX + 8.43 kg/s CH4 bleed → ORP preburner → LOX turbopump (35,000 RPM)
154.3 kg/s CH4 + 74.1 kg/s LOX bleed → FRP preburner → CH4 turbopump (50,000 RPM)
Both exhausts at 800K → Main chamber 350 bar → 3,533K combustion → 3,560 m/s exit

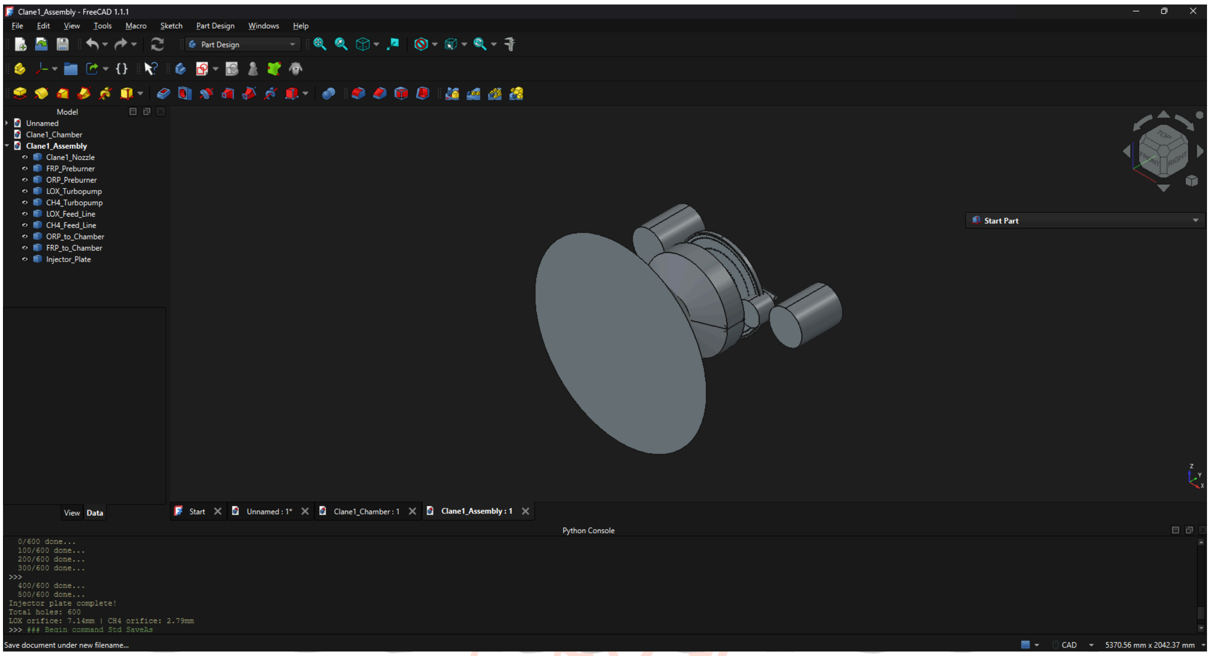
7. 3D CAD MODEL

Built in FreeCAD 1.1.1 using Python-scripted parametric approach.

All dimensions derived from thermodynamic analysis in Section 5.

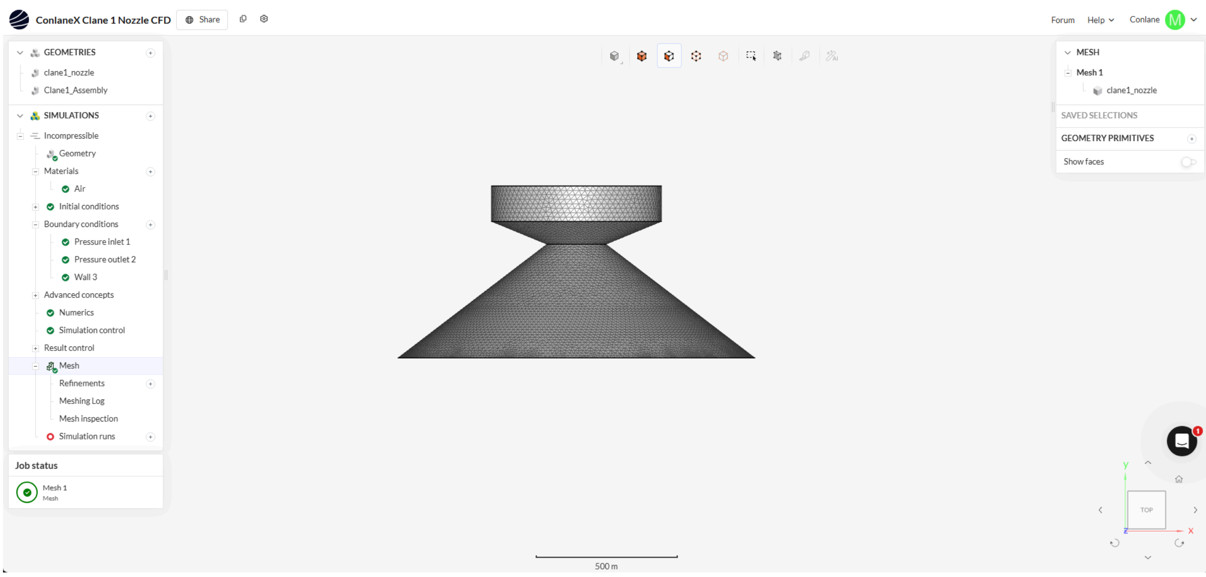
Components: Chamber + nozzle | Throat (212mm) | Bell nozzle exit (1,257mm)
FRP + ORP preburners | LOX + CH4 turbopumps | Feed lines
Injector plate — 600 coaxial elements, LOX 7.14mm / CH4 2.79mm

CAD files: github.com/memorareach-code/conlanex-clane1



8. CFD SIMULATION

Tool: SimScale + OpenFOAM (rhoCentralFoam compressible solver)
Geometry: clane1_nozzle.stl from FreeCAD
Inlet: 350 bar, 3,533K | Outlet: 1 bar | Walls: no-slip adiabatic
Mesh: 229k-344k cells complete
Status: Compressible solver run in progress
OpenFOAM case files in GitHub repository



9. COST ANALYSIS

Prototype 1 unit: $315,000
10-unit production: $195,300 per unit
50-unit production: $132,300 per unit
Raptor 3 (est.): $1,000,000 per unit

Savings drivers: India LPBF printing | Monolithic chamber | India CNC labour | IN-SPACe TAF grant

10. DEVELOPMENT ROADMAP

Phase 1 — Design (COMPLETE): Physics + CAD + CFD | Rs 0
Phase 2 — Simulation (Month 1-3): CFD + FEA | Rs 0
Phase 3 — Register (Month 3-4): Conlane Corp Pvt Ltd + IN-SPACe NGSE | Rs 15,000
Phase 4 — Grant (Month 4-5): IN-SPACe TAF up to Rs 25 Cr | Rs 0
Phase 5 — Sub-scale (Year 2): 1/10 scale hot fire 3s | Rs 2-5 Cr
Phase 6 — Full scale (Year 3-5): Full manufacture + qualification | Rs 80-150 Cr

11. ABOUT CONLANEX

ConlaneX (Conlane Corporation Pvt Ltd)
Bengaluru, Karnataka, India

Mission: Building the machines that leave Earth.

Founder: Chandan D V. Designed Clane 1 from first principles at age 17 using open-source tools and zero external capital.

GitHub: github.com/memorareach-code/conlanex-clane1
Instagram: @conlanex

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